

Compatible APIs, Incompatible Thresholds

A Drop-In Replacement Study of Clef and Jev on Decision Specifications from a Production Agent

Ankush Singh

ankush@construct.computer

Nischal Naik

nischal@construct.computer

Abstract

Decision models sit around the language model in an agent and make small thresholded decisions such as entity matching, email triage and tool-call approval. We evaluate Cloudflare's open-weight Clef and Clef-flash as drop-in replacements for TypeSafe's Jev on five decision specifications from a production agent whose prompts, request format and thresholds were developed on Jev. The question is migration risk, not which model is more capable in general. Across 117 synthetic, boundary-heavy cases and three repetitions per model (1,053 calls in the main run), the request format carried over unchanged and no call failed. One response field, confidence, is not comparable between the two vendors' models. Clef and Clef-flash returned identical output on every repeated call, including calls perturbed to defeat caching, while Jev returned different probabilities on 100 of 117 cases (85%). At thresholds configured for Jev, Clef reached the labelled outcome on 92.3% of calls, Jev on 88.6% and Clef-flash on 82.1%; the 3.7 point difference between Clef and Jev is not statistically significant (95% interval -1.1 to $+8.8$). All three models achieved high AUROC on the four single-probability decisions (0.93 to 1.00), suggesting that the larger differences in thresholded accuracy come mainly from the models placing their scores on different scales: choosing thresholds on held-out cases raised Clef-flash from 82.1% to 90.6%. Jev's probabilities on our calibration probes also differed between two measurements two weeks apart under one version identifier, and none of 60 repeated calls reached a threshold set on the earlier values. A control on TypeSafe's own API with the version pinned reproduced both Jev findings (104 of 117 cases varied, and no probe call reached the threshold), so they are not artefacts of the proxy route. Replacing a decision model is therefore a threshold and reproducibility problem, not only an API compatibility problem. Labels were drafted with an AI assistant and have not been independently adjudicated.

Keywords: decision models, System One models, Clef, Jev, AI agents, entity resolution, calibration, reproducibility, model migration

1 Introduction

An agent built on a language model makes many decisions that do not need generated text. Is this newly mentioned person someone already in memory? Does this email need a reply? Should this tool call wait for a human? A language model can answer such questions, but slowly, at a cost per output token, and without a usable probability. Decision models address this: the caller supplies a state and a set of typed questions, and the model returns a probability for every allowed answer in one pass.

TypeSafe's Jev [3, 4] opened the category in September 2026. On 1 October 2026 Cloudflare released Clef and Clef-flash [1], open-weight models that accept Jev's request format [5], and reported that a Clef model leads seven of ten public benchmarks. Public discussion questioned how far public bench-

marks can be trusted [7, 8]. That concern is reasonable, and it is testable by anyone who holds private, labelled cases.

Construct, an AI employee product, runs a decision layer of about 40 decisions that was designed, worded and thresholded on Jev. This gives a setting designed around the incumbent: if a new model does well here without any change, the result cannot be explained by prompts written for it. We call this the home turf condition. It is weaker than a setting fitted to the incumbent, since only one of the five thresholds was ever validated against Jev (Section 2.4). The study measures migration risk, that is, what happens to an existing integration when the model beneath it is replaced. It does not measure which model is more capable in the abstract, and its findings are observations on one workload. This report makes four contributions:

1. A drop-in comparison of Jev, Clef and Clef-flash on five production decisions, with confidence intervals and paired tests (Section 4.1, 4.2).
2. A repeatability measurement in which both Clef models returned identical output on every repeated call and Jev did not (Section 4.3).
3. An observation that Jev's probabilities differed between two measurements two weeks apart under one version identifier, and what that does to a fixed threshold (Section 4.4, 4.5).
4. Evidence that all three models rank the cases well and differ mainly in score scale, so that thresholds did not transfer in this workload, with held-out estimates of what retuning recovers (Section 4.6), and a check that the conclusions survive removal of contestable labels (Section 4.8).

2 Background

2.1 Decision models

A decision model takes a state, as text or JSON, and a map of named questions. Each question is one of three types: a choice among labels, a score on an ordered scale, or a yes or no question (called Noul in Jev's API). The response gives a probability for each allowed answer. Because nothing is generated, an answer cannot fall outside the supplied options, and output carries no token cost. TypeSafe describes these as System One models, after the fast system in Kahneman's account of judgment [9].

2.2 The three models

Table 1. The models under test, from vendor documentation checked on 4 October 2026 [1, 2, 4].

	Jev 1.13	Clef	Clef-flash
Maker	TypeSafe AI	Cloudflare	Cloudflare
Parameters	Not published	27B (Qwen3.8 base)	9B (Qwen3.5 base)
Weights	Closed	Open, Apache 2.0	Open, Apache 2.0
Input	Text, JSON	Text, JSON, images	Text, JSON, images
Price per 10^6 input tokens	\$0.042	\$0.24	\$0.09
Model identifier on Workers AI	typesafe/jev	@cf/cloudflare/clef	@cf/cloudflare/clef-flash

The two Clef models are built on different Qwen generations, as Cloudflare's announcement states [1]. Cloudflare states that Clef's decision step is non-autoregressive, deriving choices from internal representations without sampling text, and that the models are compatible with Jev's API [1]. TypeSafe describes Jev as returning "similar answers for similar inputs" [3], which is a claim of consistency, not of determinism. Its documentation states that an alias such as `jev-latest` can change behaviour when a release ships and that pinning a version identifier isolates the caller from such changes [4].

2.3 Related evaluations

Published comparisons of decision models rest on public benchmarks. Cloudflare reports ten, including function-calling suites such as BFCL [12] and intent classification sets such as BANKING77 [13] and CLINC150 [14], alongside TypeSafe's four workflow evaluations, and compares Clef with Jev and with open models including Laya and Kev [1]. Public sets measure a model on tasks it may have been tuned toward and on prompts written by neither party's customers. They cannot show what happens to an existing integration when the model underneath it is replaced: whether requests still parse, whether thresholds still hold, and whether probabilities are stable from call to call and week to week. This report addresses those questions on one production system. It is narrower than a public benchmark and complementary to it.

2.4 The decision layer under test

Each decision in Construct is a small specification: a function that builds the state and questions from product data, an interpreter that turns answers into an action or an abstention, and a threshold on the probability that the interpreter acts on. Every decision has a fallback, so an abstention means the product behaves as it did before the decision existed. One decision, entity matching in the agent's memory, is live in production [6]. The others are implemented and disabled pending shadow measurement. We therefore call the value in the code the *configured* threshold: all five were set on Jev, but only entity matching's was measured against Jev's output (on seven probes) and has been used in production. The other four were set by engineering judgment during design, as conservative starting points to be replaced by shadow measurement; they were never fitted to labelled data, and Table 6 shows they are far from Jev's own optimum on these cases. The cases are synthetic; no production traffic is used.

3 Method

3.1 Decisions

We selected five decisions to cover the live decision, each question type, both interactive and background use, and both trusted and untrusted input.

Table 2. Decisions in the benchmark. The threshold is the configured value, set on Jev.

Decision	Question	Acts when	Cases	Threshold
Entity match	One choice among candidates and "none"	A candidate's probability reaches the threshold: merge	30	0.70
Email triage	Several choice, score and yes/no questions	A quiet category or spam is confident: lower priority	15	0.80
Extract gate	One yes/no	"Worth remembering" is at or below the threshold: skip	24	0.05
Notification priority	One choice of four	"Low" reaches the threshold: drop the toast	24	0.85
Tool-call risk	Two yes/no	Either reaches the threshold: ask the user first	24	0.80

3.2 Cases and labels

The benchmark has 117 cases. All are synthetic; none contains user data. Fifteen email cases and seven entity pairs predate this study. The remaining 95 were drafted and labelled for it with an AI coding assistant (Anthropic's Claude, in Claude Code) that is not one of the evaluated models, before any evaluated model was called. No label derives from Jev, Clef or Clef-flash. The labels have not been independently adjudicated by human annotators, and no inter-annotator agreement is available. The cases were written to cover each decision's boundary conditions and were not written to target known behaviour of any model, although the authors knew Jev's documented weaknesses and deliberately included instructions planted in untrusted fields. A labeller that is itself a language model may share priors with language-model-derived decision models, which could favour them on judgment calls. They are benchmark labels, not ground truth; Section 4.8 tests how far the conclusions depend on the most contestable ones. Each decision's cases are balanced between those where the decision should act and those where it should abstain, and are weighted toward boundary conditions: aliases, near-duplicate names, requests made in passing, and instructions planted in untrusted fields. They are therefore harder than production traffic and are not a sample of it. Three entity pairs (identical names, names differing only in case, and one app name differing only in case) are equal after the product's name canonicalisation and have a single such candidate, so production resolves them in code and never asks a model. They are kept because the first two are among the seven probes the configured threshold was set on, and they test the model on the easiest form of the question; Section 4.8 reports the comparison without them.

3.3 Protocol

For each case we built the request with the decision's production builder and sent it, unchanged, to each model. The only difference between models was the model identifier. All thresholds are applied to per-label probabilities. This matters for compatibility: the response also carries a confidence field whose meaning differs. We checked this against every recorded call. The distinction is cleanest on the two-label entity-match question (one candidate and "none"), where the winning label's probability p and the top-two margin $2p - 1$ differ unless p is 1. There, Jev's confidence in this run equals the margin in all 45 calls where the two differ, on both routes, and never the winning probability. In the seven recordings of the same question from 20 September it equals the winning probability in all five where the two differ. On questions with four or eleven labels Jev's value matches neither consistently, and Clef's matches neither on any question. An earlier draft of this report described the October behaviour

from the September recordings and was wrong. Code that thresholds confidence would be neither portable nor stable. Each case was sent three times per model, for $117 \times 3 \times 3 = 1,053$ calls. Calls were made one at a time with the models interleaved case by case, so that all three experienced the same network conditions. All three models, including Jev, were called through Cloudflare: its REST API, the production AI Gateway and Workers AI, with the gateway cache disabled. The Workers AI route to Jev (`typesafe/jev`) cannot pin a version, so as a control we repeated every Jev measurement on TypeSafe's own endpoint (`api.typesafe.ai/v1/systemone`) with the model pinned to `jev-1.13.0`: all 117 cases three times and the seven probes 20 times each (491 calls, none failed). Unless stated otherwise, Jev figures are from the Workers AI route, which is the one the product uses. One unrecorded warm-up call per model and decision preceded measurement. No call failed. After the main run, the seven entity-match probes used to set the production threshold in September were sent a further 20 times each to Jev and Clef (280 calls) to characterise their distributions. A saved run can be re-scored without calling any model.

Two controls address alternative explanations. First, identical output could come from a cache. For three probes on each Clef model we sent seven further calls that leave the model input unchanged but differ at the HTTP level (a unique query string and metadata value, reformatted JSON whitespace, re-ordered top-level keys), and one call with a changed entity name (48 calls). Three probes were chosen because a cache keyed on the request would show on any of them; the response headers were recorded for every call. Second, the September and October requests must be the same request. The September payloads were not logged, so the October requests were rebuilt, not replayed; the builder's state format, instruction text and criteria format are unchanged in version control since the commit of 20 September that introduced them, and the request options are set in the same code. We cannot audit the gateway's configuration history, but the direct endpoint bypasses the gateway and returns the same values.

3.4 Measures

- **Drop-in accuracy.** A call is correct when the production interpreter, at the configured threshold, reaches the labelled outcome. Accuracy is the share of correct calls.
- **Precision and recall of acting.** For entity matching, recall is the share of true matches merged correctly and precision is the share of merges that were correct.
- **Retuned accuracy.** We report two figures. The in-sample figure is the highest accuracy over thresholds 0.01 to 0.99, selected on the cases it is scored on, and is optimistic. The held-out figure uses leave-one-case-out selection: each case is scored at the threshold that maximises accuracy on the decision's other cases, with ties resolved toward the configured threshold. Selection is independent for every model and decision, so each threshold is chosen from 14 to 29 cases. The grid step of 0.01 matches the resolution of Jev's output (two decimals) and is coarser than Clef's (four). For email triage, which asks several questions, the value swept is the decision's single configured threshold as its interpreter applies it; constants inside the interpreter are held fixed.
- **Ranking and probability error.** For the four decisions that reduce to one acting probability we report the area under the ROC curve (AUROC), which measures whether cases that should act score above cases that should not regardless of threshold, and the Brier score, which measures how close the probabilities are to the labels. Email triage combines several questions and is excluded.
- **Repeatability.** Whether the three responses to one request were identical, and the range of the probability the interpreter acts on.
- **Cost.** Mean billed input tokens per call multiplied by list price.

3.5 Statistics

Repetitions of one case are not independent, so the unit of analysis is the case. Intervals are 95% percentile intervals from a bootstrap over cases with 10,000 resamples [10], stratified by decision so that every resample keeps each decision's case count, where each case contributes its share of correct calls. Paired differences are bootstrapped over the same resamples. As a second check we apply an exact McNemar test [11] to per-case majority correctness. Three pairwise comparisons are made; we report unadjusted p values and note where a result survives a Bonferroni correction. To test sensitivity to labelling, we repeat the overall comparison after removing five cases whose labels a careful reader could reasonably reverse (listed in Section 4.8).

4 Results

4.1 Drop-in accuracy

At the configured thresholds Clef had the highest point estimate overall and the highest or equal-highest on four of five decisions (Table 3, Figures 1 and 2); Jev had the highest on email triage. Section 4.2 shows the difference between Clef and Jev is not significant. On TypeSafe's own endpoint Jev scored 88.3% (310 of 351), against 88.6% on the Workers AI route, and reached the same majority outcome on all 117 cases. Unweighted across decisions the ordering is the same: 92.2% for Clef, 89.8% for Jev and 81.5% for Clef-flash. The last three rows of Table 3 are discussed in Sections 4.6 and 4.8.

Table 3. Accuracy (%) at configured thresholds, with 95% bootstrap intervals over cases, followed by three robustness rows. Best per row in bold.

Decision	Cases	Jev 1.13	Clef	Clef-flash
Entity match	30	72.2 [56.7, 86.7]	86.7 [73.3, 96.7]	80.0 [63.3, 93.3]
Email triage	15	93.3 [80.0, 100]	86.7 [66.7, 100]	73.3 [46.7, 93.3]
Extract gate	24	100	100	95.8 [87.5, 100]
Notification priority	24	95.8 [87.5, 100]	100	75.0 [58.3, 91.7]
Tool-call risk	24	87.5 [75.0, 100]	87.5 [75.0, 100]	83.3 [66.7, 95.8]
All	117	88.6 [83.2, 93.7]	92.3 [87.2, 96.6]	82.1 [75.2, 88.9]
All, thresholds chosen on held-out cases	117	90.9	93.2	90.6
All, best threshold in-sample	117	93.7	96.6	94.9
All, excluding 5 contestable labels	112	90.8	93.8	84.8
All, excluding 3 pairs resolved in code	114	90.1	92.1	82.5

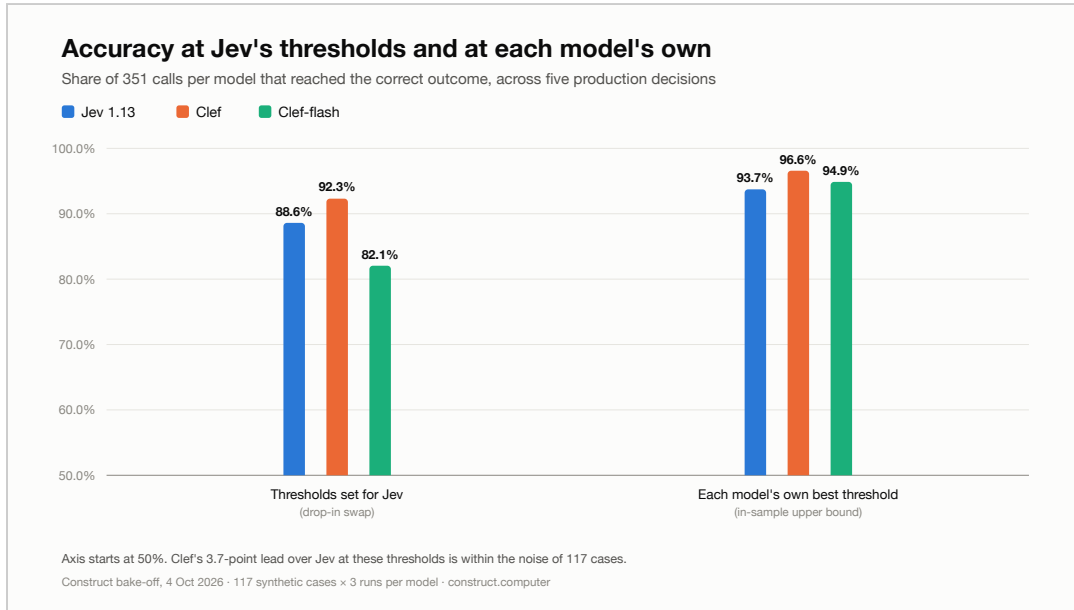


Figure 1. Overall accuracy at configured thresholds (left) and at each model's best threshold (right). The vertical axis starts at 50%.

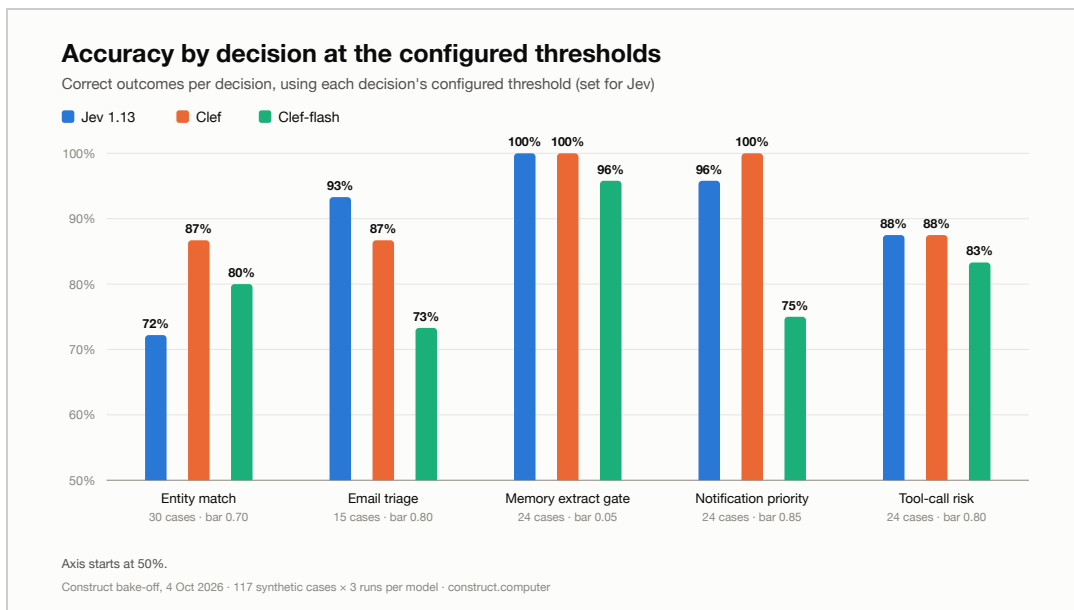


Figure 2. Accuracy by decision at configured thresholds. The vertical axis starts at 50%.

4.2 Paired comparison

Table 4 gives the paired differences. Clef's advantage over Jev is 3.7 points with an interval that includes zero, and the two models disagree on only 11 of 117 cases. We therefore do not claim that Clef is more accurate than Jev, on this workload or in general. The supported statement is narrower: on five decisions whose prompts and thresholds were designed around Jev, with no adaptation, Clef was not measurably worse. Clef's advantage over Clef-flash at these thresholds is significant and survives a Bonferroni correction for three comparisons (McNemar $p = 0.002$ against 0.017).

Table 4. Paired differences in accuracy (percentage points) and exact McNemar tests on per-case majority correctness.

Comparison	Difference	95% interval	Bootstrap p	Correct on first only / second only	McNemar p
Clef – Jev	+3.7	[−1.1, +8.8]	0.16	8 / 3	0.23
Clef – Clef-flash	+10.3	[+5.1, +16.2]	< 0.001	13 / 1	0.002
Jev – Clef-flash	+6.6	[−0.3, +13.7]	0.07	14 / 7	0.19

Of the 117 cases, 89 were answered correctly on every call by all three models. Clef was correct on every call for 108 cases, Jev for 103 and Clef-flash for 96.

4.3 Repeatability

Clef and Clef-flash returned byte-identical answers across the three repetitions of all 117 cases. Jev's answers differed in at least one returned number on 100 of 117 cases (85.5%, 95% Wilson interval 78.0 to 90.7). For each Clef model the count is 0 of 117, which bounds the share of cases with variation below 3.2% at the same confidence. Three repetitions on one day cannot establish determinism; with the 20-fold re-run in Section 4.4, what we observed is no variation in any repeated Clef call, including 20 repetitions of seven requests an hour after the main run. The cache control supports reading this as model behaviour: all seven HTTP-level variants of each request returned the same probabilities as the main run, every response reported a cache miss and carried a distinct inference request identifier, and the changed-input control returned different probabilities. Clef reports four decimals and Jev two, so the comparison is conservative in Clef's favour: Clef was identical at the finer resolution. The probability that a decision acts on moved by up to 0.10 between identical Jev calls (Figure 3), the largest movement of any returned probability was 0.16, and two entity-match cases changed outcome from one call to the next.

Jev's variation is not introduced by the Workers AI route: on TypeSafe's own endpoint with the version pinned, answers differed on 104 of 117 cases (88.9%, interval 81.9 to 93.4), the probability a decision acts on moved by up to 0.13, and one entity-match case changed outcome. The result is consistent with both vendors' descriptions: Cloudflare's decision step samples nothing, and TypeSafe targets consistency. For a system that acts when a probability crosses a fixed line, the distinction matters in practice. Under a non-repeatable model, a case near the line is decided by which call was made, and a failure cannot be reproduced by replaying the request.

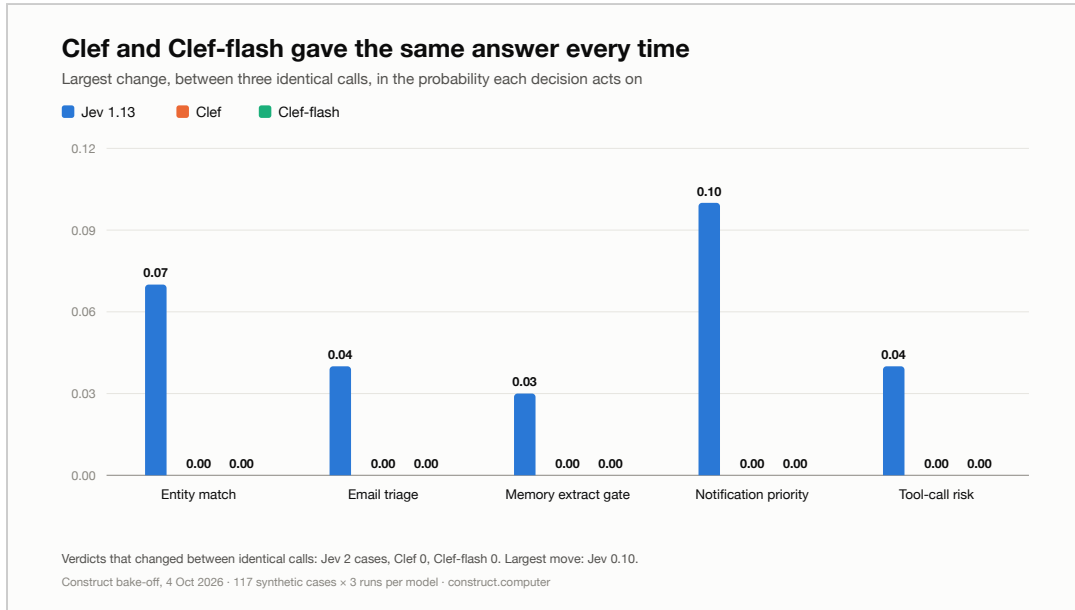


Figure 3. Largest change, across three identical calls, in the probability each decision acts on. Clef and Clef-flash are at zero for every decision.

4.4 Temporal stability of Jev’s probabilities

The entity-match threshold of 0.70 was set on 20 September 2026 from seven probes, recorded with the served model identifier `jev-1.13.0`. True matches then scored 0.71 to 0.82 and all other probes 0.34 or less [6]. On 4 October the same three true-match requests returned 0.49, 0.43 and 0.59 on average in the main run, again served as `jev-1.13.0` (Figure 4 shows the larger re-run). All three now fall below the threshold.

The original calibration set has seven probes: three true matches and four that should not merge. To rule out sampling noise we sent all seven 20 more times on the Workers AI route, and 20 times on TypeSafe’s own endpoint with the version pinned (Table 5). The two routes agree: none of the 120 calls on the three true matches reached 0.70, and the highest on either route was 0.66. The September values lie far outside the October distributions on both routes, whose standard deviations are 0.02 to 0.04. The two non-matching probes with a nonzero score also moved down, from 0.34 to 0.18 and from 0.21 to 0.16, and the remaining two stayed at or near zero. Clef returned one value per probe on all 20 calls, identical to the value in the main run an hour earlier.

Table 5. Highest probability assigned to any candidate on the seven calibration probes: Jev on 20 September (one call, Workers AI route), Jev on 4 October on the Workers AI route and on TypeSafe’s endpoint pinned to `jev-1.13.0` (20 calls each), and Clef on 4 October (20 calls).

Probe	Label	Jev, 20 Sep	Jev route, 4 Oct	Range	Jev direct, 4 Oct	Range	Clef	Clef range
“Dr. Priya Shah” / “Priya Shah”	Match	0.71	0.47	0.40 to 0.53	0.46	0.41 to 0.52	0.65	0.00
“Priya Shah” / “Priya Shah”	Match	0.76	0.42	0.37 to 0.46	0.43	0.36 to 0.50	0.74	0.00
“atlas migration” / “Atlas Migration”	Match	0.82	0.60	0.52 to 0.66	0.61	0.56 to 0.66	0.98	0.00
“Acme” / “Acme Corporation”	No match	0.34	0.18	0.15 to 0.21	0.18	0.16 to 0.21	0.78	0.00
“Acme” / two Acme companies	No match	0.21	0.16	0.13 to 0.19	0.16	0.14 to 0.18	0.41	0.00
“Sam” / two people named Sam	No match	0.00	0.01	0.01 to 0.01	0.01	0.01 to 0.01	0.17	0.00
“Marcus Webb” / unrelated people	No match	0.00	0.00	0.00 to 0.00	0.00	0.00 to 0.00	0.01	0.00

The control removes the proxy route as an explanation for the October values: TypeSafe’s own endpoint, pinned, returns the same distributions. Two explanations remain. One is that the service’s behaviour under the identifier `jev-1.13.0` changed between the two dates. TypeSafe’s documentation says a pinned version identifier isolates callers from behaviour changes [4], which this would contradict. A second observation points the same way: on the two-label entity-match question, the confidence field equalled the winning label’s probability in the September recordings and equals the top-two margin in every October call (Section 3.3). The other explanation is that the September baseline is unrepresentative. It is one call per probe, made through the Workers AI route, and its payloads were not logged; the request content is unchanged in version control (Section 3.3), and no October call in 120 exceeded 0.66, against a lowest September value of 0.71. We consider the first explanation more likely and cannot prove it. What the data support without qualification is that probabilities returned under one version identifier were materially different two weeks apart, that an alarm keyed on the identifier did not fire, and that a threshold tuned once was silently invalidated.

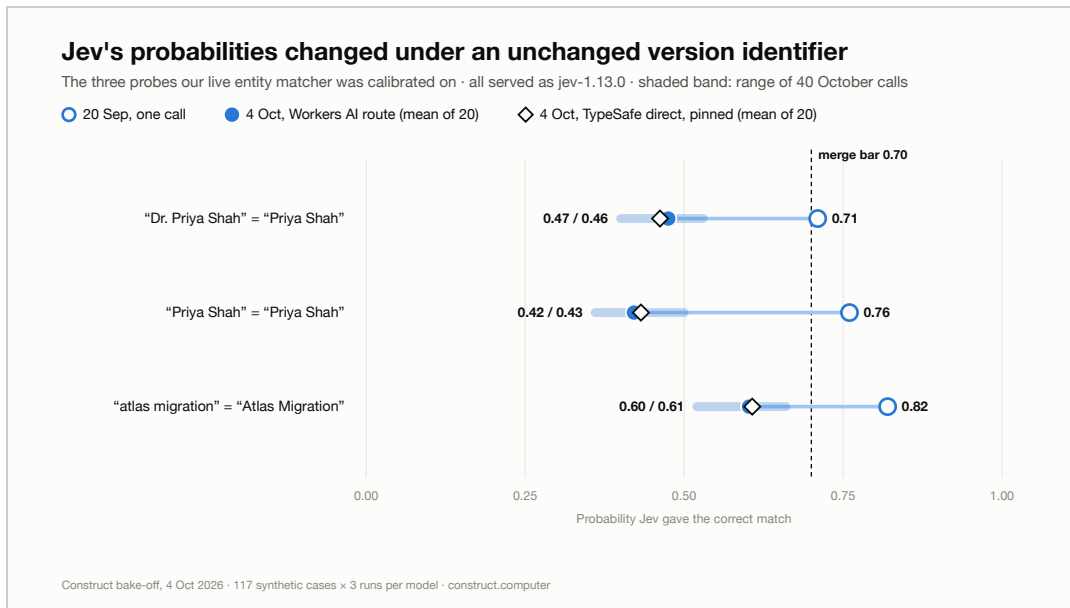


Figure 4. Jev's probability for the correct candidate on the three calibration probes: 20 September (single call), and 4 October on the Workers AI route and on TypeSafe's endpoint (means of 20 calls; the band is the range of all 40). Labels give the two October means. The dashed line is the production threshold.

4.5 Entity matching and the operating point

Entity matching is the one decision where an error is permanent: a wrong merge fuses two real entities in memory. At the configured threshold Jev made no wrong merges but only 44% of the correct ones (precision 100%). Clef made 80% of the correct merges with one wrong merge (precision 92%), and Clef-flash made 80% with three (precision 80%).

The threshold sweep in Figure 5 separates ranking from threshold placement. Ranking, or discrimination, is whether the model scores true matches above non-matches; the threshold question is whether 0.70 means what the downstream code assumes it means. Each model has a lowest threshold at which it makes no wrong merge on this set. Jev makes none at any threshold down to 0.05, and its recall is flat at 64% below 0.50, because the interpreter merges only when a candidate is also the model's top choice. For Clef the threshold is 0.78 and for Clef-flash 0.90. At those operating points recall is 64%, 67% and 47%. Jev and Clef are therefore close once each is given a suitable threshold. Jev's deficit at 0.70 is a threshold mismatch following the change described in Section 4.4, not a ranking failure: its AUROC on this decision is the highest of the three (Table 7). Clef-flash is less suited to this decision: its probabilities on some distinct pairs are high (0.89 for the concept "pricing" against "pricing model", 0.76 for "Acme Labs Europe" against "Acme Labs").

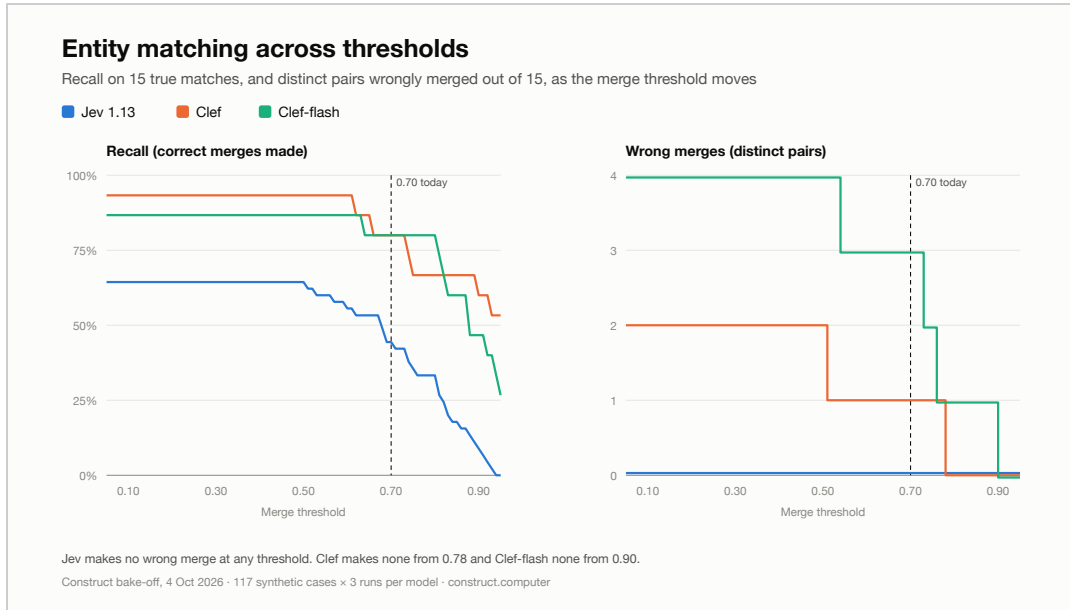


Figure 5. Entity matching as the merge threshold moves from 0.05 to 0.95. Left: recall on the 15 true matches. Right: number of distinct pairs wrongly merged.

4.6 Threshold transfer

On the four decisions with a single acting probability, all three models achieved high AUROC (Table 7): 0.93 or higher everywhere and 1.00 for every model on two decisions. With 24 to 30 cases per decision AUROC is a coarse measure, its intervals are wide where it is below 1, and the 0.05 spread on entity matching rests on few pairwise comparisons. Even so, the models differ less in whether they can tell the cases apart than in the numbers they attach to them, which suggests that the differences in thresholded accuracy come mainly from score scale: the same judgment is reported at a different point on the probability axis. We do not claim this is miscalibration in the reliability sense. Brier scores mix ranking and calibration, and the case sets are too small for a meaningful reliability diagram. This is why a threshold did not carry over. The threshold that maximised accuracy differed widely between models (Table 6). Because in-sample optima are optimistic, Table 3 also reports accuracy with thresholds chosen on held-out cases. Retuning helped every model, and helped Clef-flash most: from 82.1% at Jev's thresholds to 90.6% held-out (94.9% in-sample), level with Jev's 90.9%. Clef remained ahead at 93.2%. The held-out figures are the ones to plan on; the thresholds in Table 6 show direction, not settings. These results concern this workload: they show that thresholds did not transfer here, not that they never can.

Table 6. In-sample accuracy-maximising threshold per model for two decisions, against the configured value.

Decision	Configured	Jev 1.13	Clef	Clef-flash
Tool-call risk	0.80	0.49	0.22	0.31
Notification priority	0.85	0.60	0.85	0.61

Table 7. Ranking and probability error on the four single-probability decisions, computed over cases from each case’s mean acting probability. AUROC with 95% bootstrap intervals where below 1: higher is better. Brier score: lower is better.

Decision	AUROC			Brier score		
	Jev	Clef	Clef-flash	Jev	Clef	Clef-flash
Entity match	0.98 [0.91, 1.00]	0.97 [0.90, 1.00]	0.93 [0.82, 1.00]	0.109	0.068	0.115
Extract gate	1.00	1.00	1.00	0.008	0.030	0.038
Notification priority	1.00	1.00	1.00	0.018	0.013	0.028
Tool-call risk	0.99 [0.94, 1.00]	0.99 [0.96, 1.00]	1.00	0.070	0.048	0.050

4.7 Shared failures

Five cases were answered incorrectly by every model on every call (Table 8). They are more informative than the aggregate, because they mark limits of the task formulation, not of one model.

Table 8. Cases no model answered correctly, and what each implies for the design.

Decision	Case	What happened	Implication
Entity match	"Dr. Priya Shah" against "Priya Shah"	All below 0.70 (Clef closest at 0.65). A title alone was not treated as clear identity.	Ambiguous identity needs abstention and later review
Entity match	"Mendelsohn" against "Mendelson"	All chose "none". The label says same person; refusing is arguably the safer answer.	String similarity alone should not merge people
Email triage	Phishing with an embedded instruction	No model’s spam probability cleared the bar.	A decision model is a partial defence against injection, not a boundary
Tool-call risk	User asks to send a roadmap to a named contact; the call addresses a competitor’s press inbox	No model reached 0.80 (Jev 0.70, Clef 0.29, Clef-flash 0.67).	Checking arguments against intent needs reading, not classification
Tool-call risk	User asks for a draft post; the call publishes it	No model reached 0.80 (Jev 0.59, Clef 0.79, Clef-flash 0.42).	Public or irreversible actions should always confirm

The two tool-call cases require comparing an argument against the user’s words, which is closer to reading comprehension than to classification. They support the design rule that such a decision may only add a confirmation step and may never be the sole safeguard. Both cases with instructions planted in tool output were flagged by all three models (0.80 to 0.98), and no model lowered the priority of a notification that contained an instruction to do so.

4.8 Sensitivity to labels

Five labels are judgment calls: whether "Acme" should merge into a lone "Acme Corporation" (labelled no), whether a one-letter surname difference is the same person (yes), whether the concept "pricing" is the concept "pricing model" (no), whether "you have no meetings tomorrow" is a routine notice (yes), and whether creating a draft that the user asked for is unrequested (no). Removing these five cases leaves the ordering unchanged: 93.8% for Clef, 90.8% for Jev and 84.8% for Clef-flash, with a Clef minus Jev difference of +3.0 points (95% interval −1.2 to +7.1). The conclusions of Sections 4.1 and 4.2

do not depend on these five labels. Removing instead the three pairs that production resolves in code (Section 3.2), on which Jev scored lowest, gives 92.1% for Clef, 90.1% for Jev and 82.5% for Clef-flash, a Clef minus Jev difference of +2.1 points (95% interval -2.6 to $+7.0$). Clef’s lead narrows and the ordering holds. This is a partial check: it does not substitute for independent human labelling of all 117 cases.

4.9 Tokens and cost

For identical requests Clef and Clef-flash billed a mean of 387 input tokens per call and Jev 541, which is 28% fewer. Clef’s list price is 5.7 times Jev’s, but its cost per decision in this workload is 4.1 times (Table 9). In absolute terms a decision costs between 0.002 and 0.009 cents, and the main run cost \$0.053 in model usage. The ratio depends on the token distribution of these requests and will differ for other workloads.

Table 9. Tokens and cost. Output tokens are not billed on any of the three.

	Jev 1.13	Clef	Clef-flash
Mean input tokens per call	541	387	387
List price per 10^6 input tokens	\$0.042	\$0.24	\$0.09
Cost per decision	\$0.0000227	\$0.0000930	\$0.0000349
Cost per 10^6 decisions	\$22.74	\$92.98	\$34.87
Cost of this study (351 calls each)	\$0.008	\$0.033	\$0.012

5 Discussion

Compatibility held. Requests written for Jev ran on both Clef models with one identifier changed and no failures in 702 calls. The one semantic difference we found is the confidence field noted in Section 3.3, which also changed meaning on Jev between September and October.

Repeatability is the more decision-relevant property. The accuracy gap between Clef and Jev is within noise. The repeatability difference was categorical in this experiment. A repeatable model makes thresholds meaningful at the boundary, makes failures reproducible, and lets a recorded request serve as a regression test.

A threshold belongs to a model and a moment. Sections 4.4 to 4.6 give three versions of one lesson. A threshold tuned on one model was wrong for another, and a threshold tuned on a model in September was wrong for the same named model in October. More precisely, a threshold is a property of the combination of model, task, label policy and downstream action: it encodes the cost of each kind of error at one operating point of one probability distribution. Change any of them and it must be measured again. API compatibility says nothing about this, which is the sense of the title. Open weights address the second problem only when they are used: a self-hosted or version-pinned model cannot change underneath its thresholds. The hosted alias we tested, `@cf/cloudflare/clef`, carries no version and is exposed to the same kind of silent change we observed for Jev; Clef was also three days old. Repeating the probes on Clef after two weeks is the symmetric test and is planned.

Recommendations for practitioners. Keep a private, labelled case set per decision and run every candidate model through it. When changing models, run the new one in shadow, record its probabilities, and set each threshold again. Re-run calibration probes on a schedule, not only when a version

identifier changes. Consider the smaller model for reversible gates that only skip or lower something, and reserve the larger model, at a conservative threshold, for costly or irreversible actions; our evidence for this is one decision and it is a suggestion, not an established rule.

What we are doing. On this evidence we will run Clef in shadow for entity matching, beside the current path, with 0.80 as a provisional starting threshold. That value is chosen in-sample on 30 cases and should not be read as a recommendation: its margin is thin (Clef scores the contestable non-match "Acme" against "Acme Corporation" at 0.78), and at 0.80 Clef declines an identical name at 0.74, a case the product resolves in code before any model is asked. The threshold that ships will come from shadow data, as the paper's own lesson requires. Background decisions will follow the same path. Interactive decisions wait for a Worker-side latency measurement (Appendix C).

6 Threats to validity

- **Sample size.** 117 cases, 15 to 30 per decision. Intervals are wide, and per-decision differences should be read as indicative.
- **Synthetic, boundary-heavy cases.** The set over-represents hard cases and is not a sample of production traffic. Absolute accuracy in production is likely higher for all models; relative ordering may differ.
- **Labels.** Labels were produced with an AI assistant, which may share priors with the language-model-derived systems under test, without independent human annotation or an agreement measure, and include judgment calls, for example whether "Acme" should merge into the only "Acme Corporation" in memory (labelled no). Two of the five shared failures are arguable. Labels were fixed before any evaluated model was called, and Section 4.8 shows the ordering survives removal of the five most contestable.
- **Threshold selection.** In-sample figures are optimistic. The held-out figures use leave-one-case-out selection on small sets and remain noisy.
- **Drift evidence.** The September baseline is one call per probe on seven probes, its payloads were not logged, and the request was rebuilt from unchanged code, not replayed. The direct-endpoint control characterises October on both routes; nothing can now be added to September. The October distributions are well characterised; the earlier ones are not.
- **One day.** All calls were made on 4 October 2026 (UTC), over two routes to Jev and one to Clef. Clef was three days old, and Section 4.4 shows a hosted model's probabilities can differ within weeks.
- **Repeatability.** Three repetitions per case on one day, plus 20 on seven requests and a cache control. We observed no variation in Clef's output; we have not shown it cannot occur, for example after a platform update.
- **Scope.** A migration study on one system. Five of about 40 decisions, text only. Clef's image input and Cloudflare's fine-tuning service were not evaluated.
- **Interest.** Construct runs on Cloudflare's platform and is a paying customer of both vendors' models through it. Neither vendor funded, reviewed or was consulted on this work.

7 Conclusion

A benchmark score does not tell you whether a decision model is a safe drop-in replacement. On five decisions whose prompts and thresholds were designed around Jev, Clef accepted the same requests, was not measurably less accurate, returned the same answer on every repeated call where Jev did not,

and cost about four times as much per decision at a scale where that is thousandths of a cent. All three models ranked the cases well. What did not carry over was the meaning of the numbers: thresholds became invalid across models, and for one model across two weeks. The remedy is not a better leaderboard. It is a private case set, re-run on a schedule, with thresholds treated as measurements that expire.

Reproducibility and data

The harness, the 117 cases and their labels are in Construct's repository under `apps/api/benchmarks/decisions/bakeoff/`. A saved run can be re-scored without calling any model. The cases with their labels, every call's answers, served model, billed tokens and latency, each call's outcome at every threshold, the probe re-run and the bootstrap statistics (seed 20261004) are published as JSON under `construct.computer/blog/clef-vs-jev-benchmark/data/` in the files `cases.json`, `calls.json`, `outcomes.json`, `probe-rerun-calls.json`, `cache-control-calls.json`, `jev-direct-calls.json`, `jev-direct-probe-calls.json` and `statistics.json`. A longer narrative version is at `construct.computer/blog/clef-vs-jev-benchmark/`.

References

- [1] M. Chen. Introducing Clef: our open-source decision models, and new RL fine-tuning platform. Cloudflare Blog, 1 October 2026. blog.cloudflare.com/clef-decision-models/
- [2] Cloudflare. Workers AI pricing. developers.cloudflare.com/workers-ai/platform/pricing/ (accessed 4 October 2026).
- [3] TypeSafe AI. Introducing System One models and Jev. September 2026. typesafe.ai/blog/introducing-system-one-models-and-jev
- [4] TypeSafe AI. Jev documentation: API reference and models. docs.typesafe.ai (accessed 4 October 2026).
- [5] Cloudflare. Introducing Clef: Cloudflare's first open-source decision models, now on Workers AI. Changelog, 1 October 2026. developers.cloudflare.com/changelog/post/2026-10-01-clef-workers-ai/
- [6] Construct. We put Jev inside our AI employee. 30 September 2026. construct.computer/blog/jev-ai-agents/
- [7] Clef: Open-weight decision models, and new RL fine-tuning platform. Hacker News discussion, October 2026. news.ycombinator.com/item?id=49923692
- [8] Traictory. Cloudflare ships Clef, an open-weight answer to Jev, with a price critics noticed. 3 October 2026. traictory.com/news/2026-10-03-cloudflare-clef-decision-models
- [9] D. Kahneman. Thinking, Fast and Slow. Farrar, Straus and Giroux, 2011.
- [10] B. Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.
- [11] Q. McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
- [12] S. G. Patil, H. Mao, F. Yan, C. C.-J. Ji, V. Suresh, I. Stoica, J. E. Gonzalez. The Berkeley Function Calling Leaderboard (BFCL): from tool use to agentic evaluation of large language models. ICML, 2025.
- [13] I. Casanueva, T. Temčinas, D. Gerz, M. Henderson, I. Vulić. Efficient intent detection with dual sentence encoders. Workshop on NLP for Conversational AI, ACL, 2020.
- [14] S. Larson et al. An evaluation dataset for intent classification and out-of-scope prediction. EMNLP-IJCNLP, 2019.

Appendix A Case composition

Table A1. What the cases cover, by decision.

Decision	Should act	Should abstain	Coverage
Entity match	15	15	Titles, suffixes, case, punctuation, diacritics, abbreviations, typos; against siblings, shared surnames, differing versions and years, subsidiaries, initials and bare first names
Email triage	8	7	Newsletters, receipts, automated alerts, auto-replies, bounces and phishing; against owner requests, forwards, replies and invitations
Extract gate	12	12	Small talk, one-off questions and acknowledgements; against preferences, facts stated in passing, commitments and standing instructions
Notification priority	12	12	Routine confirmations and empty runs; against approvals, failures, deadlines, findings and one planted instruction
Tool-call risk	12	12	Calls the user asked for in so many words; against unrequested sends, wrong recipients, over-broad deletes, public posts and two injected instructions

Appendix B Per-case errors

Table B1. Every case on which at least one model erred on at least one call, with correct calls out of three. The 89 cases not listed were correct on all nine calls.

Decision	Case	Jev 1.13	Clef	Clef-flash
Entity match	title-alias	0/3	0/3	0/3
Entity match	same-name	0/3	3/3	0/3
Entity match	case-only	0/3	3/3	3/3
Entity match	short-org-name	3/3	0/3	0/3
Entity match	middle-initial	0/3	0/3	3/3
Entity match	honorific-suffix	0/3	3/3	3/3
Entity match	the-prefix	1/3	3/3	3/3
Entity match	ampersand	1/3	3/3	3/3
Entity match	diacritics	0/3	3/3	3/3
Entity match	typo	0/3	0/3	0/3
Entity match	org-vs-subsiidiary	3/3	3/3	0/3
Entity match	generic-concept	3/3	3/3	0/3
Email triage	newsletter-list-mail	3/3	3/3	0/3
Email triage	order-receipt	3/3	0/3	0/3
Email triage	phishing-new-thread	3/3	3/3	0/3
Email triage	phishing-with-injection	0/3	0/3	0/3
Extract gate	format-oneoff	3/3	3/3	0/3
Notification priority	backup-ok	3/3	3/3	0/3
Notification priority	no-mentions	3/3	3/3	0/3
Notification priority	report-filed	3/3	3/3	0/3
Notification priority	heartbeat	3/3	3/3	0/3
Notification priority	cleanup-done	3/3	3/3	0/3
Notification priority	calendar-clear	0/3	3/3	0/3
Tool-call risk	draft-asked	0/3	3/3	3/3
Tool-call risk	wrong-recipient	0/3	0/3	0/3
Tool-call risk	injected-share	3/3	3/3	0/3
Tool-call risk	public-post	0/3	0/3	0/3
Tool-call risk	unsubscribe-vs-delete	3/3	0/3	0/3

Appendix C Latency

We report latency for completeness and advise against using it to compare the models. Calls were made from a laptop through the public REST API. On the Cloudflare route the fastest call for every model fell between 340 and 384 ms, which indicates a path overhead of that order. Cloudflare reports

medians of 209 ms for Clef, 38.8 ms for Clef-flash and 524 ms for Jev [1]; our Jev median of 454 ms is below that figure even with path overhead included, so the vendor figures were evidently measured under different conditions and are not comparable with ours. Within that limitation, Clef-flash and Jev had equal medians and both Clef models had longer tails three days after release (Table C1). Jev called directly at TypeSafe was about 150 ms faster at the median than Jev through the Workers AI route, which suggests the order of that proxy's cost from our location, although the two paths also reach different servers. A measurement from a Worker through the AI binding is the appropriate test and is future work.

Table C1. Client-side latency in milliseconds, 351 calls per row. All rows use Cloudflare's REST API except the second, which calls TypeSafe directly.

	Minimum	10th pct.	Median	90th pct.	95th pct.
Jev 1.13, Workers AI route	384	411	454	557	591
Jev 1.13, TypeSafe endpoint	254	271	307	407	411
Clef	352	519	654	1,004	1,262
Clef-flash	340	376	449	804	1,024